

Theory Behind Simulations

Single VO2 Neuristor under Voltage and current sources

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Contents

1	Introduction	1
1.1	Current Dynamics Equation	1
1.2	Thermal Dynamics Equation	2
1.3	Resistance / Hysteresis Model	3
2	Derivation of the Current-Source Model	5
2.1	Starting Point: Original Voltage-Driven Equations	5
2.2	Assumptions and Circuit Interpretation	5
2.3	Step-by-Step Derivation for the Direct Current-Source Experiment	6
2.4	Final Continuous-Time Current-Driven Model	6
3	Discretization	7
3.1	Discretization of Current Dynamics Equation	7
3.2	Discretization of Thermal Dynamics Equation	8
3.3	Discretization of Resistance / Hysteresis Model	8
3.4	Discretization of the Current-Source Model	9
4	Fitting of the Resistance Model Based on Real Data	9
4.1	Input Data and Preprocessing	9
4.2	What We Obtain at Each Step vs What Must Be Assumed	10
4.3	Parameter Seeding and Intermediate Quantities	10
4.4	Optimization Objective and Fitting Procedure	11
4.5	Final Fitted Parameters, Quality Metrics, and Limits	13
5	Sources	14

1 Introduction

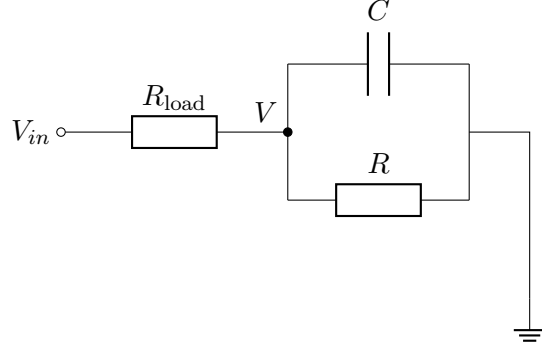


Figure 1: Single-neuristor equivalent circuit used in the voltage-driven model.

1.1 Current Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C \frac{dV_i}{dt} = \frac{V_i^{in}}{R_i^{load}} - V_i \left(\frac{1}{R_i} + \frac{1}{R_i^{load}} \right). \quad (1)$$

Single-device form used in this project:

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}), \quad (2)$$

$$\frac{dV}{dt} = \frac{V_{in} - V}{R_{series}C} - \frac{V}{R_{vo2}C}. \quad (3)$$

Variables and Units

Symbol	Meaning	Units
V_i, V	node/device voltage across VO ₂ and capacitor	V
V_i^{in}, V_{in}	applied source voltage	V
R_i, R_{vo2}	VO ₂ resistance state	Ω
R_i^{load}, R_{series}	series/load resistance	Ω
C	parasitic/effective node capacitance	F
t	time	s
$\mathcal{H}$	hysteresis internal state (branch/reversal memory)	–

Physical Interpretation

- $C dV/dt$: capacitive storage term (how fast node voltage can change).

- Source-branch current is $I_{src} = (V_{in} - V)/R_{series}$, which can be decomposed as $V_{in}/R_{series} - V/R_{series}$.
- V/R_{series} : the part of the source-branch term that opposes charging as node voltage rises.
- V/R_{vo2} : current through the VO₂ device; this term is strongly nonlinear because R_{vo2} depends on temperature and hysteresis state.

The competition between $I_{src} = (V_{in} - V)/R_{series}$ and VO₂ discharge determines oscillatory charging-switching dynamics.

1.2 Thermal Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C_{th} \frac{dT_i}{dt} = \frac{V_i^2}{R_i} - S_e(T_i - T_0) + S_c \nabla^2 T_i + \sigma \eta_i(t). \quad (4)$$

Single-device reduction ($\nabla^2 T = 0$):

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (5)$$

Variables and Units

Symbol	Meaning	Units
T_i, T	device temperature	K
t	time	s
C_{th}	thermal capacitance	J/K
V_i, V	device voltage	V
R_i, R_{vo2}	VO ₂ resistance	Ω
S_e	thermal conductance to environment/substrate	W/K
S_c	nearest-neighbor thermal coupling conductance	W/K
T_0	ambient/base temperature	K
$\nabla^2 T$	discrete thermal Laplacian (neighbor exchange term)	K
$\eta_i(t)$	Gaussian white-noise process (paper notation)	$s^{-1/2}$
σ	noise strength multiplier (paper notation)	$J s^{-1/2}$

Physical Interpretation

- $\frac{V^2}{R_{vo2}}$: Joule-heating input that increases T .
- $S_e(T - T_0)$: heat leakage to ambient that decreases T when $T > T_0$.

- $S_c \nabla^2 T$: diffusive thermal exchange with neighbors; positive when neighbors are hotter.
- $\sigma \eta_i(t)$: stochastic thermal fluctuations (paper noise term).

The thermal equation therefore balances Joule heating, environmental cooling, inter-device heat diffusion, and noise.

1.3 Resistance / Hysteresis Model

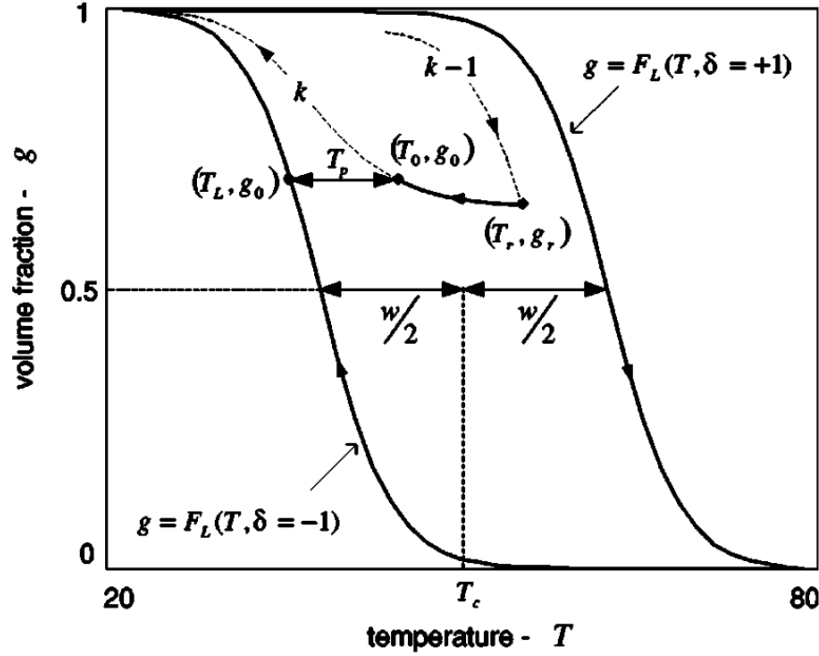


Figure 2: Reference hysteresis geometry showing major branches, reversal points, and proximity temperature shift in the volume-fraction model.

Equation(s)

Starting point from de Almeida *et al.* (2002):

$$R_s(T) = R_0 \exp\left(\frac{E_a}{k_B T}\right). \quad (6)$$

$$R(T) \approx g(T)R_s(T) + (1 - g(T))R_m \Rightarrow R(T) \approx g(T)R_s(T) + R_m. \quad (7)$$

Major-loop semiconducting fraction:

$$g_{\text{major}}(T, \delta) = \frac{1}{2} + \frac{1}{2} \tanh\left[\beta \left(\delta \frac{w}{2} + T_c - T\right)\right], \quad \delta = \begin{cases} +1, & dT/dt > 0, \\ -1, & dT/dt < 0. \end{cases} \quad (8)$$

Minor-loop memory terms:

$$T_{pr} = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_r - 1) - T_r, \quad (9)$$

$$x = \frac{T - T_r}{T_{pr}}, \quad P(x) = \frac{1}{2} (1 - \sin(\gamma x)) (1 + \tanh(\pi^2 - 2\pi x)), \quad T_p = T_{pr} P(x), \quad (10)$$

$$g(T, \mathcal{H}) = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - (T + T_p) \right) \right]. \quad (11)$$

Final resistance law used in this project:

$$R_{vo2}(T, \mathcal{H}) = R_0 \exp \left(\frac{E_a/k_B}{T} \right) g(T, \mathcal{H}) + R_m. \quad (12)$$

Variables and Units

Symbol	Meaning	Units
$R_{vo2}, R(T)$	VO ₂ device resistance	Ω
$R_s(T)$	semiconducting-branch resistance (Arrhenius)	Ω
R_m	metallic-state resistance floor	Ω
R_0	Arrhenius prefactor	Ω
E_a/k_B	activation-energy-over-Boltzmann term	K
T	device temperature (our model: Kelvin)	K
$g(T, \mathcal{H})$	semiconducting volume fraction with hysteresis memory	–
w	hysteresis width parameter	K
T_c	hysteresis center temperature	K
β	branch steepness	K ^{−1}
δ	branch sign (+1 heating, −1 cooling)	–
T_r, g_r	last reversal state (temperature, fraction)	K, –
T_{pr}	reversal-derived proximity temperature scale	K
γ	proximity-function shape parameter	–
$P(x)$	proximity function controlling minor-loop approach	–
T_p	instantaneous proximity temperature shift	K
$\mathcal{H}$	full hysteresis state ($\delta, T_r, g_r, T_{pr}, \dots$)	–
$\tilde{R}, \tilde{g}$	measured resistance and inferred fraction for fitting	$\Omega, -$

Physical Interpretation

- Eq. (6): $R_s(T)$ decreases exponentially with temperature in the insulating/semiconducting regime.
- Eq. (7): total resistance combines semiconducting contribution gR_s with a metallic floor R_m .
- Eq. (8): two major branches are selected by δ , creating the loop width w .

- Eqs. (9)–(10): reversal variables (T_r, g_r, T_{pr}) and $P(x)$ encode memory and generate minor loops.
- Eq. (12): this is the resistance passed to the electrical/thermal ODEs at every timestep.

Assumptions

- $R_s \gg R_m$ in the transition regime justifies the additive form in Eq. (7).
- Major-loop branches are represented by a symmetric tanh form around T_c .
- Hysteresis state updates only at reversal events (dT/dt sign change).

2 Derivation of the Current-Source Model

2.1 Starting Point: Original Voltage-Driven Equations

Equation(s)

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}). \quad (13)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma\eta(t). \quad (14)$$

2.2 Assumptions and Circuit Interpretation

Assumptions

- The voltage source is removed and replaced by a current source in a separate experiment.
- The source enforces an ideal programmed current waveform $I_{in}(t)$.
- An external resistor R_{series} may be present in the injection path in the physical setup.
- Under ideal current drive, adding/removing this external R_{series} does not change node-current balance, so both circuits are dynamically equivalent.
- The VO₂ branch representation is unchanged: $R_{vo2}(T, \mathcal{H})$ in parallel with C , referenced to ground.
- Thermal dynamics keeps the same Joule-heating/cooling structure as in Section 1.2.

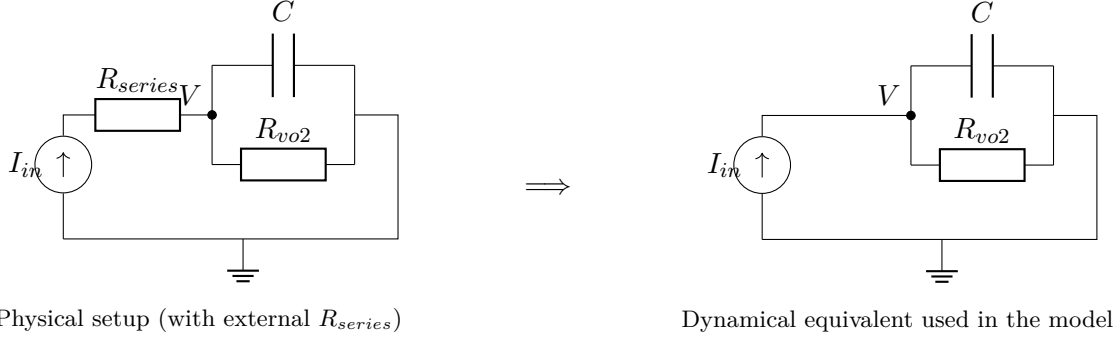


Figure 3: Current-driven setup used in practice (left) and the reduced equivalent used for derivation (right). For ideal enforced $I_{in}(t)$, the external series resistor does not alter node dynamics.

Physical Interpretation

- The imposed source current $I_{in}(t)$ sets the net injected current into node V .
- An external series resistor may drop voltage in the real wiring, but it does not change injected current when the source is ideal.
- At node V , current splits between the capacitor branch and the VO_2 branch.
- VO_2 current V/R_{vo2} remains the electrical-to-thermal coupling path through V^2/R_{vo2} .

2.3 Step-by-Step Derivation for the Direct Current-Source Experiment

Derivation Steps

$$\text{KCL at node } V : \quad I_{in}(t) = I_C + I_{vo2} = C \frac{dV}{dt} + \frac{V}{R_{vo2}}. \quad (15)$$

$$\Rightarrow \quad C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}. \quad (16)$$

This is identical whether or not an external series resistor is included in the physical current-source lead, provided the source enforces $I_{in}(t)$.

2.4 Final Continuous-Time Current-Driven Model

Equation(s)

$$C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}, \quad R_{vo2} = R(T, \mathcal{H}). \quad (17)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (18)$$

Variables and Units

Symbol	Meaning	Units
V	node voltage across VO ₂ and capacitor	V
$I_{in}(t)$	injected source current waveform	A
R_{series}	optional external series resistor in the physical setup (non-dynamical here)	Ω
R_{vo2}	hysteretic VO ₂ resistance $R(T, \mathcal{H})$	Ω
C	electrical capacitance at the node	F
T	device temperature	K
C_{th}	thermal capacitance	J/K
S_e	thermal conductance to environment	W/K
T_0	ambient/base temperature	K
σ	thermal-noise amplitude (current module convention)	$\text{W}\sqrt{\text{s}}$
$\eta(t)$	white-noise term	–
$\mathcal{H}$	hysteresis internal state	–

3 Discretization

3.1 Discretization of Current Dynamics Equation

Discrete Update Equation

Using forward Euler with time step Δt , and $R_n = R(T_n, \mathcal{H}_n)$:

$$\left. \frac{dV}{dt} \right|_n = \frac{V_{in,n} - V_n}{R_{series}C} - \frac{V_n}{R_nC}. \quad (19)$$

$$V_{n+1} = V_n + \Delta t \left. \frac{dV}{dt} \right|_n. \quad (20)$$

Equivalent compact form:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left[\frac{V_{in,n}}{R_{series}} - V_n \left(\frac{1}{R_n} + \frac{1}{R_{series}} \right) \right]. \quad (21)$$

State Update Order

1. Evaluate hysteretic resistance R_n from current thermal state T_n and hysteresis memory $\mathcal{H}_n$.
2. Compute dV/dt at step n from Eq. (19).
3. Update node voltage to V_{n+1} (explicit Euler).

3.2 Discretization of Thermal Dynamics Equation

Discrete Update Equation

Define $P_n = V_n^2/R_n$. The thermal derivative at step n is:

$$\left. \frac{dT}{dt} \right|_n = \frac{P_n - S_{env}(T_n - T_0) + S_{couple} \nabla_d^2 T_n}{C_{th}} + \xi_n. \quad (22)$$

Forward Euler update:

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (23)$$

where ∇_d^2 is the discrete Laplacian and ξ_n is optional additive temperature-rate noise.

For the simplified single-device non-deterministic case ($S_{couple} = 0$):

$$\left. \frac{dT}{dt} \right|_n = \frac{1}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \xi_n, \quad (24)$$

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (25)$$

State Update Order

1. Use electrical state V_n and resistance R_n to compute Joule power $P_n = V_n^2/R_n$.
2. Compute thermal derivative from Eq. (22) (or Eq. (24) in the reduced case).
3. Update temperature to T_{n+1} (explicit Euler).

3.3 Discretization of Resistance / Hysteresis Model

Discrete State Definitions

$$\mathcal{H}_n = \left\{ \delta_n, T_{r,n}, g_{r,n}, T_{pr,n}, \text{reversed}_n, T_n^{last} \right\}. \quad (26)$$

Given T_n and $\mathcal{H}_n$, the discrete evaluation is:

$$T_{p,n} = T_{pr,n} P \left(\frac{T_n - T_{r,n}}{T_{pr,n}} \right) \text{reversed}_n, \quad (27)$$

In implementation, a small ε regularization is used in the denominator when $|T_{pr,n}|$ is very small.

$$g_n = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta_n \frac{w}{2} + T_c - (T_n + T_{p,n}) \right) \right], \quad (28)$$

$$R_n = R_0 \exp \left(\frac{E_a/k_B}{T_n} \right) g_n + R_m. \quad (29)$$

Update Order (including reversals)

1. Compute $\Delta T_n = T_n - T_n^{last}$. If $|\Delta T_n|$ exceeds threshold and $\text{sign}(\Delta T_n) \neq \delta_n$, trigger reversal.

2. At reversal, set:

$$\delta_n \leftarrow \text{sign}(\Delta T_n), \quad g_{r,n} \leftarrow g(T_n, \mathcal{H}_n^-), \quad T_{r,n} \leftarrow T_n, \quad (30)$$

$$T_{pr,n} \leftarrow \delta_n \frac{w}{2} + T_c - \frac{1}{\beta} \text{arctanh}(2g_{r,n} - 1) - T_{r,n}, \quad \text{reversed}_n \leftarrow 1. \quad (31)$$

3. Evaluate g_n and R_n from Eqs. (28)–(29).

4. Set $T_{n+1}^{last} \leftarrow T_n$ for the next reversal check.

3.4 Discretization of the Current-Source Model

Discrete Update Equation

For the direct ideal-current model of Section 2:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left(I_{in,n} - \frac{V_n}{R_n} \right), \quad R_n = R(T_n, \mathcal{H}_n). \quad (32)$$

Thermal update in the current module (Euler-Maruyama in implementation):

$$T_{n+1} = T_n + \frac{\Delta t}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \frac{\sigma}{C_{th}} \sqrt{\Delta t} \zeta_n, \quad \zeta_n \sim \mathcal{N}(0, 1). \quad (33)$$

State Update Order

1. Evaluate R_n from $(T_n, \mathcal{H}_n)$.
2. Update V_{n+1} with Eq. (32).
3. Update T_{n+1} with Eq. (33).
4. Update hysteresis memory using (T_n, T_{n+1}) for the next step.

4 Fitting of the Resistance Model Based on Real Data

4.1 Input Data and Preprocessing

What is directly measured

The raw export provides eight columns per sample:

$$\{T, R, t, I, V_+, V_-, T_+, T_-\}$$

with units $\{\text{K}, \Omega, \text{s}, \text{A}, \text{V}, \text{V}, \text{K}, \text{K}\}$.

Table 1: Representative rows from the experimental dataset (actual values).

T (K)	R (Ω)	t (s)	I (A)	V_+ (V)	V_- (V)	T_+ (K)	T_- (K)
369.8230	17.6701	161.6042	2.0×10^{-6}	5.42×10^{-5}	-1.65×10^{-5}	369.818	369.828
338.0850	30.9270	399.6742	2.0×10^{-6}	6.23×10^{-5}	-6.14×10^{-5}	338.212	337.958
305.7635	3288.1260	642.2995	2.0×10^{-6}	6.55×10^{-3}	-6.61×10^{-3}	305.892	305.635
289.8545	4625.3550	824.9370	2.0×10^{-6}	9.23×10^{-3}	-9.27×10^{-3}	289.845	289.864
305.2965	3485.7080	939.4225	2.0×10^{-6}	6.98×10^{-3}	-6.96×10^{-3}	305.168	305.425
370.1625	17.5121	1424.8130	2.0×10^{-6}	5.63×10^{-5}	-1.38×10^{-5}	370.036	370.289

What is cleaned/derived

Preprocessing is:

1. Remove the units row (the row containing K, Ohm, sec, ...).
2. Convert all columns to numeric and drop invalid rows.
3. Keep rows with valid Temperature and Resistance; sort by Time.
4. If $\text{median}(T) < 200$, interpret T as $^{\circ}\text{C}$ and convert to K.

Then we compute directional masks from

$$\Delta T_n = T_n - T_{n-1}, \quad \text{cooling if } \Delta T_n < 0, \quad \text{heating if } \Delta T_n > 0. \quad (34)$$

For this dataset after cleaning:

$$N = 264, \quad T \in [289.8545, 370.1625] \text{ K}, \quad R \in [17.3765, 4625.355] \Omega.$$

4.2 What We Obtain at Each Step vs What Must Be Assumed

Table 2: Given vs computed vs unknown quantities in the resistance fit pipeline.

Class	Quantities
Given by experiment	$T(t)$, $R(t)$, time stamps, and auxiliary channels (I , V_+ , V_- , T_+ , T_-).
Computed directly from data	ΔT_n , cooling/heating masks, sweep turning point, and dataset bounds ($T_{\min}$, $T_{\max}$, $R_{\min}$, $R_{\max}$).
Unknown after preprocessing (must be fitted or swept)	R_0 , E_a/k_B , R_m , w , T_c , β , start branch (insulator or metal), and optionally γ if enough minor-loop information exists.

4.3 Parameter Seeding and Intermediate Quantities

Step-by-step outputs

The implemented seeding path is:

1. **Seed** ($R_0, E_a/k_B, R_m$) **from extremes.** A robust low-temperature window is selected; then R_m is scanned and $\ln(R - R_m)$ is linearly regressed versus $1/T$. For this dataset:

$$R_0^{seed} = 10.9606, \quad (E_a/k_B)^{seed} = 1753.57 \text{ K}, \quad R_m^{seed} = 0.8688 \Omega.$$

2. **Build seed semiconducting fraction.**

$$g_n^{seed} = \text{clip}\left(\frac{R_n - R_m^{seed}}{R_0^{seed} \exp((E_a/k_B)^{seed}/T_n)}, 0, 1\right). \quad (35)$$

3. **Seed major-loop geometry** (w, T_c). Cooling/heating temperatures closest to $g^{seed} = 0.5$ are used:

$$w^{seed} = 5.804 \text{ K}, \quad T_c^{seed} = 331.235 \text{ K}.$$

4. **Seed start branch.** From the first non-zero ΔT : `start_branch_hint = metal`.

4.4 Optimization Objective and Fitting Procedure

Objective function

The optimizer minimizes, for parameter vector x and branch b :

$$\mathcal{L}(x, b) = \text{RMSE}_{\log R}(x, b) + \lambda_g \text{RMSE}_g(x, b) + \mathcal{P}_\gamma, \quad \lambda_g = 0.2. \quad (36)$$

with

$$\text{RMSE}_{\log R} = \sqrt{\frac{1}{N} \sum_{n=1}^N [\log_{10} R_n^{model}(x, b) - \log_{10} R_n^{data}]^2}, \quad (37)$$

$$\text{RMSE}_g = \sqrt{\frac{1}{N} \sum_{n=1}^N [g_n^{model}(x, b) - g_n^{exp}(x)]^2}, \quad (38)$$

$$g_n^{exp}(x) = \text{clip}\left(\frac{R_n^{data} - R_m(x)}{R_0(x) \exp((E_a/k_B)(x)/T_n)}, 0, 1\right). \quad (39)$$

When `--fit-gamma` is active:

$$\mathcal{P}_\gamma = 0.01 |\gamma - \gamma_{\text{fixed}}|,$$

otherwise $\mathcal{P}_\gamma = 0$.

Meaning of Terms and Acronyms

- **RMSE:** *Root Mean Squared Error*. It is the square root of the mean squared residual.
- $\text{RMSE}_{\log R}$: RMSE computed on $\log_{10}(R)$, so multiplicative resistance mismatches are penalized more uniformly across decades.

- RMSE_g : RMSE between model and reconstructed semiconducting fraction g , used to keep hysteresis-shape consistency.
- λ_g : scalar weight that controls the influence of the g -consistency term relative to resistance fitting ($\lambda_g = 0.2$ in this work).
- $\mathcal{P}_\gamma$: optional regularization term for γ , used only when `--fit-gamma` is enabled.
- $\text{clip}(u, 0, 1)$: saturation/projection operator to enforce physical bounds for a fraction:

$$\text{clip}(u, 0, 1) = \begin{cases} 0, & u < 0, \\ u, & 0 \leq u \leq 1, \\ 1, & u > 1. \end{cases}$$

- x : optimization vector containing fitted parameters (e.g., $R_0, E_a/k_B, R_m, w, T_c, \beta$, optionally γ).
- b : initial hysteresis branch choice (`insulator` or `metal`) used when evaluating a candidate x .

This optimizer is a custom implementation. Methodologically, it is a bounded derivative-free search (random exploration + coordinate local refinement), conceptually related to direct-search/pattern-search methods (see Sources 3–5) and standard nonlinear optimization practice (Source 6).

Algorithm

1. **Bounded initialization:** define lower/upper bounds for each component of x , then build initial candidates from (i) baseline values and (ii) data-driven seeds.
2. **Branch screening:** evaluate candidates on both initial branches (`insulator`, `metal`) and keep the currently best pair (x, b) .
3. **Global exploration (random search):** for many iterations, propose new points by either: (a) Gaussian perturbation around current best, or (b) uniform draw in the full box. Every proposal is projected back into bounds componentwise by clipping.
4. **Local refinement (coordinate search):** update one coordinate at a time with $\pm$ step moves; accept only improving moves. If no coordinate improves, shrink step size and continue.
5. **Final branch check:** with the best x , evaluate both branches once more and keep the better one.
6. **Report:** output final parameters and RMSE metrics (overall, cooling-only, heating-only).

4.5 Final Fitted Parameters, Quality Metrics, and Limits

Results

For the fitted case documented here:

Table 3: Fitted resistance/hysteresis parameters.

Parameter	Value
R_0	0.5297358678
E_a/k_B (K)	2658.0234292
R_m (Ω)	18.1978381368
w (K)	6.9003303359
T_c (K)	333.5701800799
β	0.2969914699
γ	0.9562696820 (fixed in this run)
Start branch	metal

Table 4: Fit quality metrics ($\log_{10} R$ RMSE).

Metric	Value
Overall RMSE	0.0375554
Cooling RMSE	0.0343180
Heating RMSE	0.0406742

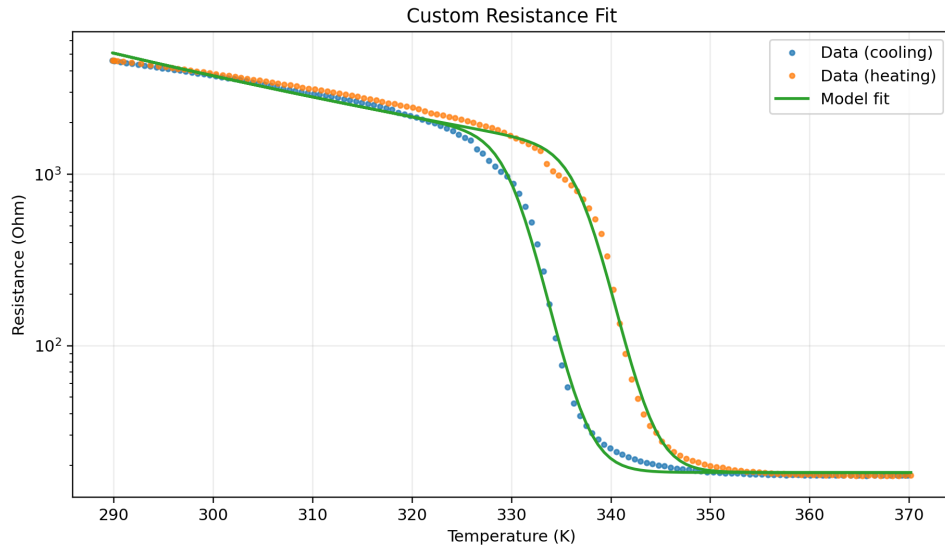


Figure 4: Measured $R(T)$ points and fitted model for the dataset used in Section 4.

Limitations

At the end of this procedure, the following still require additional identification work:

- **Minor-loop fidelity:** one major cooling-heating cycle does not fully constrain history parameters; dedicated reversal experiments are needed.
- **γ identifiability:** with major-loop data only, γ is weakly identifiable, so it is fixed unless additional data is provided.
- **Parameter uniqueness:** different parameter tuples can produce very similar $\log R(T)$ errors; confidence intervals are not yet computed.
- **Not part of this fit:** electrical/thermal dynamical parameters (C , C_{th} , S_e , source terms) must be calibrated separately from transient simulations.

5 Sources

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